POWER PULSE

Economic / Variable Load Factors Important Formulae

Notes by Balaji Sir

1 Average load and load factors

1. $$P_{avg} = \frac{\text{Area under load curve (kWh)}}{\text{Total time in hrs}}$$

average load

2. $$PLF = \frac{P_{avg}}{P_{max}} = \frac{P_{avg} \times T}{P_{max} \times T} = \frac{\text{Area under load curve (kWh)}}{\text{Rectangular area corresponding to } P_{max}}$$

Plant Load Factor: ideal PLF = 1; practical PLF < 1

3. $$PCF = \frac{P_{avg}}{P_c} = \frac{\text{Energy produced in total hrs}}{\text{Energy able to produce as per } P_c \text{ in total hrs}}$$

Plant Capacity Factor (P_c = plant capacity): ideal PCF = 1; practical PCF < 1

4. $$PUF = \frac{\text{Energy produced in used hrs}}{\text{Energy able to produce as per } P_c \text{ in used hrs}}$$

Plant Use Factor: ideal PUF = 1; practical PUF < 1

2 Utilization and reserve

5. $$U_f = \frac{P_{max}}{P_c}$$

Utilization Factor: ideal U_f = 1; practical U_f < 1

6. $$R_c = P_c - P_{max} = \frac{P_{max}\,(PLF - PCF)}{PCF}$$

Reserve Capacity = plant capacity − maximum demand

3 Demand and diversity

7. $$D_f = \frac{P_{max}}{\text{Sum of connected load}}$$

Demand Factor: ideal D_f = 1; practical D_f < 1

8. $$Div_f = \frac{\text{Sum of individual max. demand}}{P_{max} \text{ of station}}$$

Diversity Factor: practical Div_f > 1